Kexin Wei

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Robotics & medical-devices engineer specializing in motion control, surgical imaging, and sensor integration, with a track record of clinically validated, animal-trial-tested systems.

Experience

Research Engineer — Mar 2025 — Present
Imperial Global Singapore. — *Singapore*

- **Real-Time Control**: Implemented and tuned a real-time closed-loop glucose-insulin patient controller (MATLAB → Python) within a multi-container simulation stack.
- **Systems**: Built a Flask + Vue.js application for real-time simulation and cyberattack visualization.
- **Security**: Researching agentic automated penetration testing for medical cyber-physical systems.
- **Infrastructure**: Dockerized deployment and administers lab workstations, servers, and container infrastructure.

Software Team Lead — Mar 2024 — Jan 2025
Creative Medtech Solutions Pte.Ltd. — *Singapore*

- **Robot**: Engineered robotics control systems including kinematics optimization for 4-DoF robot and control interfaces.
- **Image**: Developed 4D DICOM processing and US/MRI registration algorithms for medical image alignment.
- **CyberSecurity**: Enhanced security through encryption protocols, SQL injection protection, and streamlined Qt based GUI.
- **Lead**: Led a team of 6 engineers: Git workflow, Scrum adoption, GoogleTest integration, and code-review standards.

Research and Development Engineer — Jun 2022 — Mar 2024
Creative Medtech Solutions Pte.Ltd. — *Singapore*

- **Robot**: Designed a motion-tracking algorithm for a 6-DoF robot arm integrating an A*STAR CNN, verified in vivo across pig and dog trials.
- **Sensor**: Integrated electromagnetic sensor system in the HIFU treatment software for 3D visualization of the ultrasound probe.
- **Signal Processing**: Created HIFU power-calibration and kidney respiratory-tracking algorithms, verified in ex-vivo and animal trials.
- **Patent**: Filed 3 patents in medical algorithm design: HIFU calibration, respiratory tracking, and time synchronization.

Education

M.Eng in Medical Robotics and AI — Aug 2020 — May 2022
National University of Singapore, Singapore — *GPA: 4.63/5.00*

- Thesis: Deep Reinforcement Learning (DRL) for Robot-assisted Surgical Training. Code.
- TA for ME2143E Motor Characteristics and ME5405 Machine Vision.

B.Eng in Mechanical Engineering — Sep 2014 — Jul 2019
Tongji University, China — *GPA: 4.61/5.00*

Awards & Honors

AI	Hackathon “Runner Up” Prize at bio LLMs hosted by 4Catalyzer (2023)
Mathematics Modeling	2nd Prize in CUMCM (2018); Honorable Mention in MCM/ICM (2018)
Scholarship	3rd Class Award (2018); 1st Class Award (2016); 3rd Class Award (2015)

Skills

Robotics	ROS, 6-DoF motion control, kinematics, EM/US tracking, VTK, OpenCV, DICOM
Programming	C++, Python, MATLAB, Qt, Linux, Git, Docker, SQL, Flask, Vue.js
AI/ML	PyTorch, TensorFlow, **Deep Learning** (CV, CNN, DRL), segmentation (SAM/MedSAM)
CAD/CAE	Solidworks, AutoCAD
Language	**English** (C1), **German** (C1), **Mandarin** (native)